



UNIVERSITY of
RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF INFORMATION SYSTEMS

A Final Year Project Report

**Design and implement hospital to hospital
patient data exchange platform**

*A project submitted in partial fulfilment of the requirements for the
degree of Bachelor of Science with Honours in Information Systems*

By

222011744 – IZERE MPUHWE Bede

222003541 – IZABAYO Claudien

222010573 – ISHIMWE KWIZERA Irene

Supervised by Mrs Kellen SUMWIZA

May, 2026

Abstract

Background

Rwanda's national referral hospitals, including CHUK and King Faisal Hospital, use computerized hospital information systems and relational databases to store patient data. Despite this digital capacity, there is no mechanism for direct electronic exchange of patient information between these hospitals. During referrals, critical data is often transferred using paper-based documents or manually generated reports, leading to fragmented information, delayed care, repeated tests, and potential medical errors. This reveals a gap between digital data storage and effective digital data exchange within the healthcare system.

Methods

The project began with an assessment of existing patient data management systems, database structures, and referral workflows at CHUK and King Faisal Hospital to identify interoperability challenges. Based on the findings, a centralized system architecture was designed to function as an interoperability layer connecting the independent hospital databases without replacing them. The platform enables standardized exchange of key patient information, including demographics, medical history, laboratory results, imaging reports, and treatment details.

Results

The implemented platform successfully enabled secure, real-time sharing of patient data between CHUK and King Faisal Hospital during referrals. Authorized clinicians were able to access relevant patient records from the originating hospital's database through the centralized exchange layer. The system reduced reliance on paper-based data transfer, and enhanced continuity of care by providing timely access to comprehensive patient information.

Conclusion

The project demonstrates that a centralized interoperability platform can effectively bridge independent hospital databases and support secure inter-hospital patient data exchange without replacing existing systems. This solution improves clinical decision-making, and continuity of care. The platform provides a scalable model that can be extended to other healthcare institutions in Rwanda, contributing to the country's broader digital health transformation.

Table of Contents

Abstract.....	I
List of figures.....	III
CHAPTER I: GENERAL INTRODUCTION	1
1.1 Background	1
1.2 Problem statement	2
1.3 Project objectives	3
1.3.1 General Objective	3
1.3.2 Specific Objectives	4
1.3.3 Project Questions and Hypothesis.....	Error! Bookmark not defined.
1.4 Scope of the project.....	4
1.4.1 Institutionally	4
1.4.2 Functionally	4
1.4.3 Technically	5
1.5 Project schedule (Gantt chat)	8
1.6 Organization of the project	9
CHAPTER II: LITERATURE REVIEW.....	10
2.1 Introduction.....	10
2.2 Existing similar works/systems	10
2.3 Gaps Identified	14
2.4 Expected Project Outcomes	15
CHAPTER III: METHODOLOGY	17
3.1 Introduction.....	17
3.2 Data Collection Techniques.....	17
3.2.1 Primary Data Collection.....	17

3.2.2 Secondary Data Collection	17
3.3 Software Development Methodology	18
3.4 System Analysis and Design Methods	19
3.4.1 System Analysis	Error! Bookmark not defined.
3.4.2 System Design.....	Error! Bookmark not defined.
3.5 System Diagrams.....	Error! Bookmark not defined.
3.5.1 System Architecture Diagram.....	Error! Bookmark not defined.
3.5.2 Use Case Diagram	Error! Bookmark not defined.
3.5.3 Data Flow Diagram (Level 0).....	Error! Bookmark not defined.

REFERENCES

List of figures

Figure 1.5 Project schedule or Gantt chart	8
Figure 3.5.1 System Architecture Diagram.....	Error! Bookmark not defined.
Figure 3.5.2 Use Case Diagram.....	Error! Bookmark not defined.
Figure 3.5.3 Data Flow Diagram (Level 0)	Error! Bookmark not defined.

CHAPTER I: GENERAL INTRODUCTION

1.1 Background

The healthcare system in Rwanda is organized in a structured referral hierarchy that includes health posts, health centers, district hospitals, provincial hospitals, and national referral hospitals. At the highest level of this hierarchy are national referral hospitals such as Centre Hospitalier Universitaire de Kigali (CHUK) and King Faisal Hospital, which provide specialized and tertiary healthcare services to patients referred from across the country.

Healthcare governance and policy formulation are managed by the Ministry of Health, while implementation and technical oversight are largely handled by institutions such as the Rwanda Biomedical Centre (RBC). In recent years, Rwanda has demonstrated strong commitment to digital health transformation through the adoption of electronic medical records (EMRs), hospital information systems (HIS), and national health information platforms.

At the hospital level, institutions such as CHUK and King Faisal Hospital use computerized hospital information systems supported by relational database management systems (RDBMS) to store and manage patient data. These systems typically rely on centralized relational databases (such as SQL Server and Oracle) to store structured patient information including demographics, diagnoses, laboratory results, prescriptions, and clinical notes. Each hospital maintains its own database instance hosted within its internal infrastructure or secured data centers.

While these database systems are effective for internal hospital operations, they are designed to function independently. The databases at CHUK and King Faisal Hospital are not directly connected, synchronized, or interoperable. As a result, patient data is stored in separate silos, with each hospital maintaining its own version of patient records using different database schemas, data models, and system configurations.

In the current referral workflow, when patients are transferred between hospitals (particularly between CHUK and King Faisal), patient information is commonly exchanged through paper-based referral letters, printed laboratory results, CDs for imaging data, or manually generated

reports. Even where digital records exist, access to these records is restricted to the originating hospital's database system. This limits the ability of receiving hospitals to electronically retrieve complete and up-to-date patient information.

Given that CHUK and King Faisal Hospital frequently manage complex cases that require continuity of care across facilities, the lack of a shared or interoperable database mechanism presents significant operational challenges. Clinicians at receiving hospitals often begin treatment without access to prior medical history stored in the originating hospital's database, despite that data existing electronically.

This situation highlights a critical gap between digital data availability and digital data exchange within Rwanda's healthcare system. Although hospitals use modern database systems to store patient information, there is no centralized or federated data exchange platform that enables secure, standardized, and real-time sharing of patient data between hospitals.

Therefore, the design and implementation of a centralized hospital-to-hospital patient data exchange digital platform that will be capable of interfacing with existing relational databases at CHUK and King Faisal Hospital is essential. Such platform would act as an interoperability layer, allowing hospitals (CHUK and King Faisal) to securely exchange patient data without replacing their existing database systems, by improving referral efficiency, continuity of care, and overall healthcare service delivery in Rwanda.

1.2 Problem statement

Despite Rwanda's significant progress in adopting digital health systems, the healthcare sector continues to face challenges related to interoperability and data sharing between hospitals.

National referral hospitals such as CHUK and King Faisal Hospital use computerized hospital information systems supported by relational database management systems to store patient data electronically. However, these database systems operate independently and are designed primarily to support internal hospital operations.

Currently, there is no centralized or standardized digital mechanism that enables CHUK and King Faisal Hospital to securely exchange patient data directly from their existing databases. As

a result, patient information remains fragmented across separate hospital systems, even when patients are referred between these two major referral hospitals.

During inter-hospital referrals, patient data is often transferred through paper-based referral forms, printed reports, or manually generated summaries, despite the fact that complete medical records already exist in electronic databases at the originating hospital. This disconnect between digital data storage and digital data exchange leads to several challenges, including delayed referrals, incomplete patient information at receiving hospitals, and repeated diagnostic tests.

Furthermore, clinicians at receiving hospitals frequently lack timely access to critical patient information such as medical history, laboratory results, imaging reports, allergies, and ongoing treatments. This can compromise clinical decision-making, particularly in emergency cases, and increases the risk of medical errors.

Although CHUK and King Faisal Hospital utilize modern database systems capable of storing structured patient data, the lack of an integration layer or central exchange platform prevents these systems from communicating with one another. This highlights a critical gap in Rwanda's healthcare digital infrastructure: the absence of a secure hospital-to-hospital patient data exchange platform that can integrate existing databases without replacing them.

1.3 Project objectives

1.3.1 General Objective

To design and implement a centralized digital platform that enables secure, real-time hospital-to-hospital patient data exchange between CHUK and King Faisal Hospital.

1.3.2 Specific Objectives

The specific objectives of this project are:

- ✓ To design a secure API-based interface that connects the independent relational databases of CHUK and King Faisal Hospital without replacing or disrupting their existing systems.
- ✓ To implement a centralized data exchange layer that enables authorized clinicians to access shared patient records in real time during inter-hospital referrals.
- ✓ To develop a role-based access control mechanism that ensures only authorized hospital staff can submit referrals and retrieve patient information through the platform.
- ✓ To implement standardized data exchange mechanisms that support consistent and structured sharing of patient demographics, medical history, laboratory results, and imaging reports between the two hospitals.
- ✓ To validate the platform through functional testing to confirm that patient data is exchanged accurately, securely, and without loss between CHUK and King Faisal Hospital.

1.4 Scope of the project

This project focuses on the design and implementation of a centralized hospital-to-hospital patient data exchange digital platform with a clearly defined institutional, functional, technical, and operational scope.

1.4.1 Institutionally

the study will be conducted at Centre Hospitalier Universitaire de Kigali (CHUK) and King Faisal Hospital, these two hospitals are selected as national referral hospitals that frequently exchange patient referrals, The platform will be implemented as a pilot system intended to demonstrate feasibility of inter-hospital data exchange.

1.4.2 Functionally

the system will support the electronic exchange of selected patient data required during referrals, including: Patient demographic information, medical history and diagnoses, Laboratory test results, Imaging and investigation reports, Medication and treatment information and the

platform will enable: Secure submission of referral requests, Authorized access to shared patient records by clinicians, Viewing of patient data during inter-hospital transfers.

1.4.3 Technically

the platform will be designed to integrate with existing hospital database systems at CHUK and King Faisal Hospital, Existing hospital information systems and databases will not be replaced, A centralized exchange layer will be implemented to facilitate data sharing between the two hospitals.

Primary users will include: Clinicians involved in patient referrals, Authorized hospital staff managing referrals

1.5 Motivation

The motivation for this project arises from a direct and observable gap in Rwanda's healthcare system. During hospital visits and field observations conducted at CHUK and King Faisal Hospital, it became evident that clinicians receiving referred patients frequently lacked access to those patients' prior medical records. Despite the fact that both hospitals maintain sophisticated electronic database systems, patient information was still being transferred through printed documents, handwritten referral letters, and physical CDs containing imaging data.

This situation is not merely an administrative inconvenience. In emergency and complex clinical cases, a receiving clinician who lacks access to a patient's medical history, current medications, allergies, and prior laboratory results is forced to make critical decisions with incomplete information. This increases the risk of medical errors, leads to repeated diagnostic tests that add cost and delay to treatment, and ultimately compromises the quality of care delivered to patients who are already in a vulnerable condition.

As students of Information Systems with a focus on practical solutions to real-world problems, we were motivated by the conviction that the technology required to solve this problem already exists. The databases are in place. The networks are available. What is missing is the integration layer that connects them. This project is driven by the desire to demonstrate that a well-designed,

lightweight interoperability platform can bridge that gap without requiring hospitals to replace or significantly modify their existing systems.

Beyond the immediate clinical context, this project is motivated by Rwanda's national commitment to digital health transformation. The Rwanda Biomedical Centre's Digital Health Strategic Plan 2020–2024 explicitly identifies inter-hospital data exchange as a national priority [1]. Contributing a working solution to this priority, even at the prototype level, represents a meaningful academic and practical achievement.

1.6 Significance of the Study

The proposed hospital-to-hospital patient data exchange platform carries significant implications for multiple stakeholders within Rwanda's healthcare and technology sectors.

For patients, the most direct benefit is improved continuity of care. When a patient is referred from CHUK to King Faisal Hospital or vice versa, the receiving clinician will have immediate access to their complete medical history, reducing the time required to begin appropriate treatment and lowering the risk of adverse events caused by incomplete information.

For clinicians, the platform eliminates the frustration and clinical risk associated with paper-based referral processes. Rather than waiting for physical documents to arrive or making treatment decisions based on incomplete summaries, authorized clinicians can query the exchange platform and retrieve structured, up-to-date patient records in real time. This supports better clinical decision-making, particularly in time-sensitive situations.

For hospital administrators, the platform provides an auditable record of all data exchange transactions through its audit logging functionality. This supports accountability, compliance with Rwanda's data protection law, and the monitoring of referral volumes and patterns between institutions.

For Rwanda's digital health ecosystem, this project contributes a scalable and replicable model for inter-hospital data exchange. The centralized exchange layer architecture used in this platform can be extended to connect additional hospitals within Rwanda's national referral network, supporting the broader goals of the Rwanda Biomedical Centre's digital health strategy

[1]. It also demonstrates that healthcare interoperability is achievable within a resource-constrained environment without the need for expensive proprietary systems.

For the academic community, this project adds to the limited body of research on practical health information exchange implementations in Sub-Saharan Africa, an area where published literature remains scarce despite the urgency of the problem.

1.7 Research Questions

This project is guided by the following research questions:

1. How can a centralized interoperability platform be designed to connect the independent relational databases of CHUK and King Faisal Hospital without replacing their existing systems?
2. What technical mechanisms are required to enable authorized clinicians to securely access and share patient records in real time during inter-hospital referrals?
3. How can standardized data exchange formats ensure that clinical information including demographics, laboratory results, and imaging reports is transferred consistently between the two hospitals?
4. What role-based access control structure is necessary to protect patient data confidentiality while allowing legitimate clinical access across institutional boundaries?
5. To what extent does the implemented platform reduce dependence on paper-based referral processes and improve the completeness of patient information available at the receiving hospital?

1.5 Project schedule (Gantt chat)

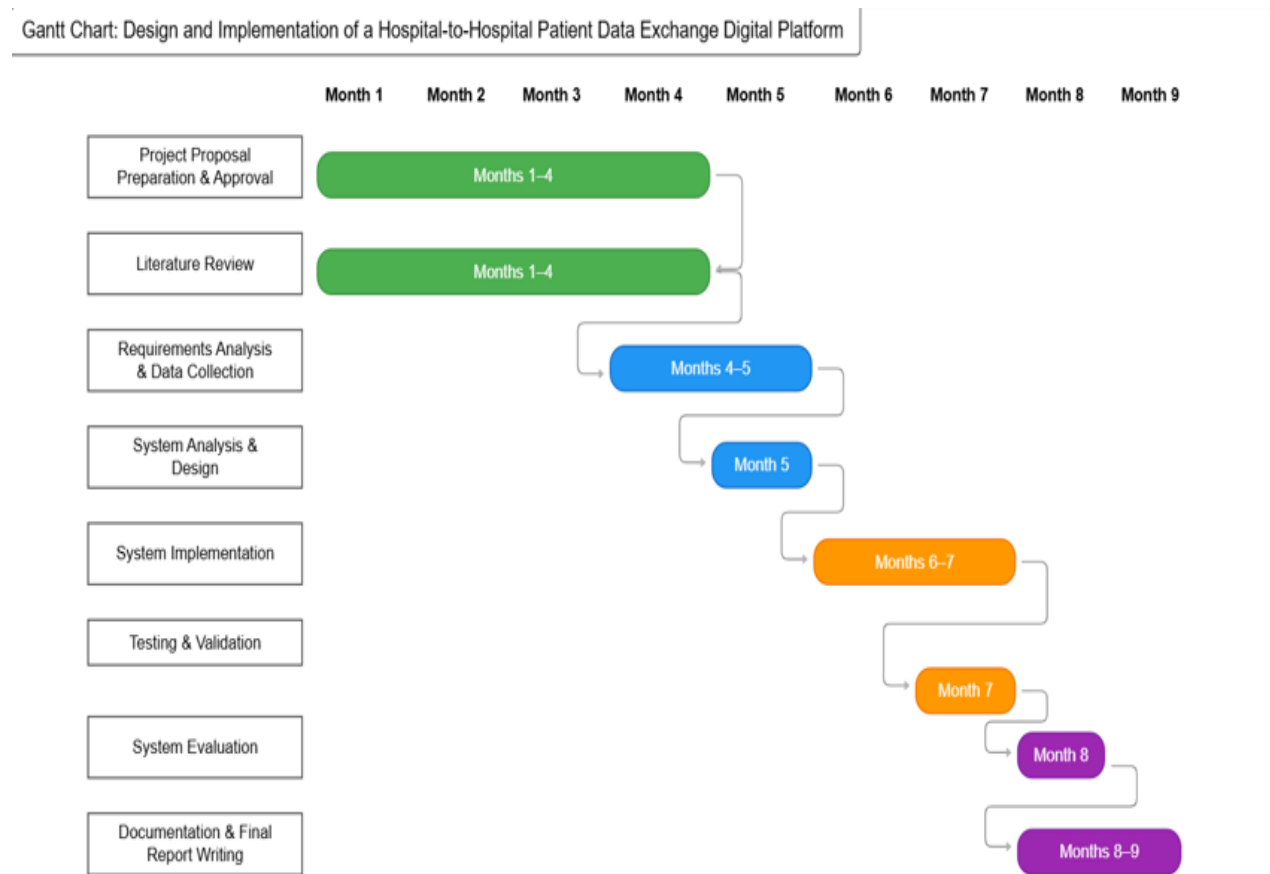


Figure 1.5 Project schedule or Gantt chart

1.6 Organization of the project

The is divided and organized in three chapters:

- ✓ **CHAPTER I:** deals with the background of project, problem statement, Research objectives, Scope of project and project organization.
- ✓ **CHAPTER II:** deals with literature review introduction, existing similar works/systems, gaps identified and expected project outcomes.
- ✓ **CHAPTER III:** deals with methodology will be used, introduction, data collection techniques, software development methodology and system analysis and design method.

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

This chapter presents the theoretical foundations and a critical review of existing literature relevant to the design and implementation of a hospital-to-hospital patient data exchange platform. It examines key concepts including health information exchange, interoperability standards, electronic medical records, role-based access control, and API-based integration. It then reviews existing systems implemented globally and within Africa, identifies challenges specific to developing countries like Rwanda, and highlights the gaps in current knowledge that this project seeks to address.

2.2 Related Work, Findings, and Existing Systems

2.2.1 Theoretical Concepts

Health Information Exchange

Health Information Exchange (HIE) refers to the electronic mobilization of clinical information across different organizations and information systems while preserving the meaning of the information being exchanged. According to the Healthcare Information and Management Systems Society, HIE enables healthcare professionals to access and securely share patient medical information electronically, improving the speed, quality, safety, and cost of patient care [2].

Three primary HIE models exist. The directed exchange model transmits information directly from one provider to another through a secure channel, functioning similarly to encrypted email. The query-based exchange model allows providers to search for and request patient information from other institutions when and as needed. The consumer-mediated model gives patients direct control over the flow of their own health information between providers [3]. The platform proposed in this project adopts the query-based model, as it allows clinicians at the receiving hospital to pull patient records from the originating hospital's database at the moment of referral.

Interoperability Standards: HL7 and FHIR

Interoperability in healthcare refers to the ability of different information systems, devices, and applications to access, exchange, integrate, and cooperatively use data in a coordinated manner

[4]. Achieving this requires agreed-upon standards that define how data is structured and communicated.

Health Level Seven International (HL7) provides a set of internationally recognized standards for transferring clinical and administrative data between healthcare software systems. HL7 Version 2 has been widely adopted for messaging between hospital systems, particularly for laboratory results, admission records, and patient demographics [5].

Fast Healthcare Interoperability Resources (FHIR), the most recent standard developed by HL7, uses modern web-based technologies including RESTful APIs and JSON data formats to enable real-time exchange of health records. FHIR is increasingly adopted in national digital health strategies because it is easier to implement and compatible with modern web architectures [4]. The proposed platform draws on FHIR principles in the logical mapping of patient data fields transmitted between hospital databases.

Electronic Medical Records

An Electronic Medical Record (EMR) is a digital version of a patient's paper chart containing the medical and treatment history managed within one institution. Unlike paper records, EMRs allow structured clinical data to be analyzed and, where interoperability is enabled, shared across healthcare settings [6]. At CHUK and King Faisal Hospital, EMRs are stored within relational database management systems including SQL Server and Oracle. While effective for internal operations, these systems are not designed to communicate with one another, creating data silos that hinder continuity of care during referrals [1].

Role-Based Access Control

Role-Based Access Control (RBAC) is a security mechanism that restricts system access based on roles assigned to individual users within an organization. In healthcare information systems, RBAC ensures that only authorized personnel access sensitive patient data, supporting compliance with data privacy requirements [7]. In the proposed platform, RBAC governs three distinct user roles: referring clinician, receiving clinician, and system administrator. Each role carries specific permissions for submitting referrals, viewing patient records, and managing system configurations respectively.

API-Based Data Integration

An Application Programming Interface (API) is a set of protocols that allows different software applications to communicate with each other. In healthcare data exchange, APIs serve as the technical bridge between independent hospital databases, enabling data to be requested and retrieved without physically merging or replacing existing systems [8]. RESTful APIs, which use standard HTTP methods, are the most widely adopted approach for modern healthcare interoperability due to their compatibility with a wide range of database platforms and programming environments. The proposed platform uses a RESTful API architecture implemented as a centralized exchange layer connecting the databases of CHUK and King Faisal Hospital.

2.2.2 The eHealth Exchange United States

The eHealth Exchange is one of the largest health information exchange networks in the United States. Originally established under the Office of the National Coordinator for Health IT, it now operates as an independent non-profit network connecting federal agencies and thousands of private healthcare providers [9].

The network operates on a query-based model, allowing participating hospitals to retrieve patient records from other facilities using standardized HL7 and Integrating the Healthcare Enterprise (IHE) protocols. It supports both directed exchange for routine referrals and push notifications for care transitions. Research has demonstrated that the eHealth Exchange significantly reduced duplicate diagnostic testing and improved clinical decision-making at receiving facilities [10].

The eHealth Exchange serves as the primary architectural benchmark for this project. Its centralized interoperability layer, standardized data formats, and role-based access mechanisms directly informed the design of the proposed platform.

2.2.3 OpenMRS Global Deployment

OpenMRS is an open-source electronic medical record system widely deployed in low- and middle-income countries including Rwanda, Kenya, Uganda, and South Africa. Designed to be

modular and adaptable to different clinical environments, it provides a flexible foundation for managing patient records at the facility level [6].

While OpenMRS excels at managing patient data within a single institution, it does not natively support cross-institutional patient data exchange. Research initiatives exploring the addition of HL7 FHIR interoperability modules to OpenMRS have demonstrated feasibility but remain largely experimental in Sub-Saharan African deployments [11]. This limitation is directly relevant to the Rwanda context, where OpenMRS is deployed in district hospitals that cannot exchange patient records with national referral hospital systems.

2.2.4 Kenya's Digital Health Platform

Kenya has made significant progress in national health data management through the Kenya Health Information System (KHIS), which uses DHIS2 to connect community health units, health centers, and referral hospitals under a unified reporting framework [12]. However, Kenya's system focuses primarily on aggregate public health surveillance data rather than real-time patient-level exchange during individual referrals. Facility-level patient records remain largely managed in independent systems that do not communicate directly with one another. This demonstrates that even countries with advanced national health information infrastructure continue to face the specific challenge of real-time inter-facility patient data exchange at the point of care.

2.2.5 India's Ayushman Bharat Digital Mission

India's Ayushman Bharat Digital Mission (ABDM), launched in 2021, represents one of the most ambitious national digital health interoperability frameworks in the developing world. It assigns every citizen a unique Health ID and implements a federated architecture through which approved health providers can access patient records across institutions using a consent-based model [8].

The ABDM's principle of authorized, structured, and auditable data exchange between independent institutions is directly applicable to the design of the proposed CHUK and King Faisal Hospital platform. While Rwanda does not yet support citizen-facing digital health

identities at scale, the underlying architecture of a centralized exchange layer governing inter-institutional access provides a valuable design reference.

2.2.6 Rwanda's Digital Health Strategy

Rwanda's Ministry of Health and the Rwanda Biomedical Centre have developed a Digital Health Strategic Plan for the period 2020 to 2024, which explicitly identifies interoperability between hospital information systems as a national priority [1]. The strategy acknowledges that while individual hospitals have adopted electronic health systems, the absence of data exchange mechanisms between facilities remains a barrier to seamless patient care.

The RBC has supported OpenMRS deployment in district hospitals and has worked toward standardizing data elements across facilities. However, at the national referral hospital level, CHUK and King Faisal Hospital operate on proprietary relational database platforms that have not been integrated with any national exchange framework [1]. The proposed platform responds directly to this unresolved gap.

2.3 Identified Gaps and Challenges

2.3.1 Absence of Inter-Hospital Data Exchange in Rwanda

The most critical gap identified in the literature is the complete absence of a real-time patient-level data exchange mechanism between Rwanda's national referral hospitals. While Rwanda's digital health strategy identifies this as a priority, no published work describes an implemented solution connecting CHUK and King Faisal Hospital's existing database systems [1]. The proposed platform addresses this gap directly.

2.3.2 Infrastructure Limitations in Developing Countries

A recurring challenge in HIE implementation across developing countries is the inconsistency of digital infrastructure. Unreliable internet connectivity, limited server capacity, and unstable power supply create barriers to real-time exchange that are largely absent in high-income country contexts [13]. The proposed platform must therefore be designed to handle degraded network conditions without data loss or corruption.

2.3.3 Heterogeneous Database Systems

Hospitals in developing countries often adopt database systems independently, resulting in environments where different facilities use different vendors, schemas, and data models. This heterogeneity makes direct database-to-database integration technically complex without a standardized intermediary layer [7]. The situation at CHUK and King Faisal Hospital, each

maintaining a distinct relational schema, exemplifies this challenge and justifies the centralized exchange layer approach adopted in this project.

2.3.4 Data Privacy and Security Requirements

Patient data is among the most sensitive categories of personal information. Rwanda's Law on Personal Data Protection requires that personal data be processed lawfully, with appropriate technical measures in place to ensure its security [1]. Implementing robust authentication, end-to-end encryption, and comprehensive audit logging is therefore both a technical requirement and a legal obligation for the proposed platform.

2.3.5 Resistance to Change and Digital Literacy

Studies across Sub-Saharan Africa consistently identify clinician resistance to new digital workflows as a significant barrier to healthcare system adoption [13]. Clinicians accustomed to paper-based referral processes may be reluctant to adopt new platforms if they are perceived as complex or unreliable. User interface design and practical training are therefore as important as the underlying technical architecture in determining whether the platform is used effectively in practice.

2.4 Way Forward

Based on the gaps and challenges identified, the following directions guide the proposed platform design.

A centralized interoperability layer offers the most practical path forward for connecting the independent databases of CHUK and King Faisal Hospital. Unlike a federated model, which requires significant shared infrastructure investment, or a point-to-point integration, which does not scale beyond two institutions, a centralized exchange layer acts as a single intermediary that both hospitals connect to independently. This approach is lower in cost, easier to extend to additional institutions, and consistent with the architectural model validated by the eHealth Exchange [9].

Adopting data exchange standards inspired by HL7 FHIR ensures that the platform maps patient data fields in a consistent and structured format, reducing the risk of information loss or misinterpretation during transfer [4]. Even where full FHIR compliance is not implemented at the prototype stage, applying FHIR's logical data models for demographics, clinical

observations, and diagnostic reports provides a standards-aligned foundation for future development.

Implementing JWT-based authentication and TLS encryption for all data transmissions directly addresses the privacy and security requirements identified in Rwanda's data protection law. Combined with RBAC and comprehensive audit logging, these mechanisms ensure that the platform meets the legal and ethical standards required for handling patient health information [7].

Finally, designing a simple and intuitive clinician interface reduces the barrier to adoption identified in the literature. A web-based dashboard that mirrors familiar referral workflows, supplemented by practical training for clinical staff, maximizes the likelihood that the platform delivers its intended clinical benefits in practice [2].

CHAPTER III: METHODOLOGY

3.1 Introduction

This chapter presents the methodology applied in the design and implementation of the hospital-to-hospital patient data exchange platform for CHUK and King Faisal Hospital. It describes the research approach used to understand the problem, the data collection techniques employed to gather requirements from real stakeholders, the tools selected for development, the nature of the system output, and the detailed system requirements that guided the design and implementation process.

The methodology was structured to ensure that every design decision was grounded in real-world evidence gathered from the hospitals, supported by established software development practices, and aligned with the specific interoperability challenges identified during the requirements analysis phase.

3.2 Data Collection Methods and Techniques

To ensure that the platform addresses genuine clinical and operational needs at CHUK and King Faisal Hospital, a combination of primary and secondary data collection methods was employed. This multi-method approach mirrors best practices in information systems research and ensures that requirements are validated from multiple perspectives before system design begins [1].

3.2.1 Interviews

Semi-structured interviews were conducted with clinical and administrative staff at both CHUK and King Faisal Hospital. The interview participants included referring clinicians, receiving clinicians and IT staff responsible for managing the hospitals' internal information systems. A total of eight interviews were conducted across both institutions.

The interviews were designed to capture three categories of information. First, the current referral workflow was documented in detail to know how a clinician initiates a referral, what information is currently sent with the patient, what format that information takes, and how long the process takes from initiation to the receiving hospital having usable information.

Second, the limitations of the existing process were explored to know specifically, what information is most frequently missing or incomplete when a patient arrives at the receiving hospital, and what clinical consequences have resulted from those gaps.

Third, stakeholder expectations for a digital solution were gathered to know what features clinicians would consider essential, what security and privacy concerns they had, and what level of technical complexity they were comfortable navigating.

Key findings from the interviews included the following. Clinicians at both hospitals confirmed that paper-based referral letters were the primary method of transferring patient information, and that these letters frequently omitted current medication lists, recent laboratory results, and

imaging reports. IT staff confirmed that both hospitals' database systems were technically capable of supporting API connections but that no integration layer currently existed between them.

3.2.2 Document Review

A systematic review of existing documentation was conducted to supplement the primary data collected through interviews. Documents reviewed included the current referral form templates used at both hospitals, the Rwanda Biomedical Centre's Digital Health Strategic Plan 2020–2024 [1], the Rwanda Ministry of Health's National eHealth Policy [2], published research on health information exchange in sub-Saharan Africa [14], and technical documentation for the HL7 FHIR interoperability standard [15].

The document review provided the regulatory and technical context necessary to ensure that the proposed platform aligns with national digital health policy and internationally recognised data exchange standards. It also confirmed that no existing platform currently connects the database systems of CHUK and King Faisal Hospital, validating the originality and necessity of the proposed solution.

3.3 Software Development Methodology

The Waterfall methodology was adopted as the primary development framework, with the recognition that the project produces a functional prototype rather than a production-ready deployment. The methodology was applied through clearly defined sequential phases such as requirements analysis, system design, implementation, testing, and evaluation with formal review points between each phase. Requirements were gathered comprehensively before design began, minimising the primary risk associated with the Waterfall model. The prototype scope was defined precisely to match the validated requirements, covering the core referral submission, patient record retrieval, authentication, and audit logging functions.

The methodology was implemented through the following six phases, each of which corresponds to sections within this report.

- ❖ **Requirements Analysis** involved gathering system requirements from clinicians and hospital staff at CHUK and King Faisal Hospital through the data collection methods described in Section 3.2. Functional and non-functional requirements were documented and validated with stakeholders before design began.

- ❖ **System Design** translated the validated requirements into a technical architecture. This phase produced the system architecture diagram, data flow diagram, database schema, and user interface wireframes described in Sections 3.5 and 3.6.
- ❖ **Implementation** involved the actual development of the platform, including the construction of the centralized exchange layer, the RESTful API connecting both hospital databases, the role-based authentication system, and the clinician web interface. The technologies used are described in Section 3.4.
- ❖ **Testing and Validation** verified the correctness and reliability of the implemented system. Testing was conducted at three levels such as unit testing of individual components, integration testing of the full data exchange flow between the simulated CHUK and KFH databases.
- ❖ **Evaluation** assessed the system against the project objectives defined in Chapter I to determine whether each specific objective had been achieved and to what extent the platform improved on the current paper-based referral process.
- ❖ **Documentation** produced the technical and user documentation necessary to support the system's future development and potential deployment, including the final project report.

3.4 Tools to Be Used

3.4.1 Hardware Tools

The hardware environment used for the development and testing of the platform is described in Table 3.1

Table 3.1: Hardware tools used in system development and testing

Component	Specification	Purpose
Development laptop	Intel Core i7, 16GB RAM, 512GB SSD	Platform development and local testing
Local server (simulated)	Vite dev server for frontend and Express.js for backend on localhost	Simulating the central exchange layer
CHUK database simulation	MySQL on local machine	Simulating CHUK's internal relational DB
KFH database simulation	Oracle database on local machine	Simulating KFH's internal relational DB

Network environment	University of Rwanda campus LAN	Simulating inter-hospital network connectivity
Testing devices	Two laptops on same LAN	Simulating CHUK and KFH clinician endpoints

3.4.2 Software Tools

The software stack was selected based on four criteria such as compatibility with the relational database systems used at CHUK and King Faisal Hospital, support for RESTful API development, availability of robust authentication libraries, and accessibility for future developers who may extend the platform. Table 3.2 describes the software tools used.

Table 3.2: Software tools used in system development

Tool / Technology	Version / Notes	Purpose
React	18.2.0	Frontend UI library
TypeScript	^5.x	Typed JavaScript for frontend and backend
Vite	8.0.0	Frontend dev server and build tool
Tailwind CSS	3.4.3	Utility-first styling
Node.js	(backend runtime)	Server-side runtime
Express	4.18.2	Backend REST API framework
Prisma	5.8.0	ORM for central database access
PostgreSQL	postgresql data source	Central application database
MySQL	external DB source	CHUK patient/external hospital records
Oracle Database XE	localhost:1521/XEPDB1	KFH external patient/hospital data source
JSON Web Tokens	jsonwebtoken 9.0.2	Authentication token handling
bcryptjs	2.4.3	Password hashing
dotenv	16.6.1	Environment variable config
CORS	2.8.5	Backend cross-origin support
Git / GitHub	—	Version control
Visual Studio Code	—	Primary code editor

- ❖ **React + TypeScript + Vite + Tailwind** are used for the frontend because they enable a modern, fast, type-safe clinician UI with lightweight development and styling.
- ❖ **Node + Express** are used for the backend REST API because they provide a simple, flexible server framework that integrates cleanly with JavaScript/TypeScript.
- ❖ **Prisma** is used as the ORM for database modeling and migration management, making it easier to manage schema changes and queries.
- ❖ **PostgreSQL** is the central exchange database because it is reliable, open source, and works well with Prisma.
- ❖ **MySQL** is used for CHUK's external data source because it is an open-source relational DB with broad compatibility.
- ❖ **Oracle XE** is used for KFH's external data source to mirror the hospital's Oracle-based system.
- ❖ **jsonwebtoken** and **bcryptjs** are used for authentication and secure password handling, while **dotenv** manages environment configuration.

3.5 Prototype or Simulation

The output of this project is a functional prototype. It is not a simulation, and it is not a production-ready deployed system.

A simulation would model the behavior of the system without actually implementing the underlying components for example, using a flowchart tool or a network modeling application to represent how data would flow between hospitals without writing any real code or creating any real database connections. The proposed platform goes significantly beyond this: it is a working software system with a real web interface, a real API, real database connections, and real authentication logic.

A production-ready system would be fully deployed on the live hospital networks of CHUK and King Faisal Hospital, connected to their actual patient databases, and used by real clinical staff in real referral scenarios. The proposed platform does not reach this level because connecting to live hospital patient databases requires institutional approval, data governance agreements, and security audits that are outside the scope of a final year academic project.

The prototype implements all core functional requirements in a controlled environment using simulated hospital databases that mirror the structure of the real systems at CHUK and KFH. It demonstrates the technical feasibility of the proposed architecture and provides a working foundation that could be extended to a production deployment with the addition of the appropriate institutional approvals and security certifications.

3.6 System Requirements

3.6.1 Functional Requirements

Functional requirements define what the system must do like the specific behaviors and functions it must be capable of performing. Table 3.3 lists the functional requirements derived from the stakeholder interviews and document review.

Table 3.3: Functional requirements of the platform

No	Requirement	Priority	Source
1	The system shall allow an authorized clinician at CHUK to submit a digital referral request containing patient demographics, diagnosis, and referral reason	High	Interviews
2	The system shall retrieve the referred patient's medical history, laboratory results, imaging reports, current medications, and allergy information from the originating hospital's database	High	Questionnaires
3	The system shall transmit retrieved patient data to the receiving hospital's authorized clinician in real time	High	Interviews
4	The system shall authenticate all users using username and password credentials before granting access to any patient data	High	Document review
5	The system shall enforce role-based access control, granting referring clinicians, receiving clinicians, and system administrators different levels of access	High	Interviews
6	The system shall generate and maintain an audit log recording every data access event, including the user identity, timestamp, and data accessed	High	Document review
7	The system shall allow authorized clinicians to view the status of submitted referrals (pending, accepted, rejected)	Medium	Interviews

8	The system shall notify the receiving clinician via the platform dashboard when a new referral has been submitted	Medium	Questionnaires
9	The system shall allow a system administrator to manage user accounts, assign roles, and revoke access	Medium	Observation
10	The system shall generate a referral summary report that can be viewed and printed by the receiving clinician	Low	Observation

3.6.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that the system must satisfy. Table 3.4 specifies the non-functional requirements with measurable thresholds where applicable.

Table 3.4: Non-functional requirements with specifications and units

No	Requirement	Specification	Unit / Threshold
1	Response time; The system shall retrieve and display patient records within an acceptable time after a clinician submits a query	Maximum 5 seconds under normal network conditions	Seconds
2	Availability; The platform shall be operational during hospital working hours without unplanned interruptions	Minimum 99% uptime during 07:00–22:00	Percentage
3	Security; All data transmitted between the clinician interface, the exchange layer, and the hospital databases shall be encrypted	TLS 1.2 or higher encryption on all connections	Protocol version
4	Authentication; JWT tokens issued upon login shall expire after a defined inactivity period to prevent unauthorized session reuse	Token expiry: 60 minutes of inactivity	Minutes
5	Data completeness; The system shall transfer a minimum defined set of patient data fields with every referral	At least 8 mandatory data fields per referral including name, DOB, diagnosis, medications, allergies, last lab result date, referring clinician, and referral reason	Number of fields

6	Usability; Clinical staff with no prior training shall be able to complete a referral submission within an acceptable time	First-use task completion: under 10 minutes	Minutes
7	Audit retention; Audit log records shall be retained for a minimum period in compliance with health data regulations	Minimum 5 years retention	Years
8	Data integrity; Patient records transmitted through the exchange layer shall arrive at the receiving system without loss or corruption	Zero data loss on transmission verified through checksum validation	Data loss rate
9	Compatibility; The exchange layer API shall be compatible with both MySQL and Oracle database query interfaces	Verified connection to both MySQL and Oracle schemas in testing	Database types

3.7 System Analysis

3.7.1 Analysis of the Existing System

The existing patient data transfer process at CHUK and King Faisal Hospital is entirely paper-based, despite both institutions maintaining sophisticated electronic database systems. The key limitations of the existing system are as follows.

Patient information is only available to the originating hospital. When a patient is referred, the receiving hospital has no access to the electronic records stored in the originating hospital's database system. The receiving clinician is entirely dependent on whatever documents the patient physically carries.

Information is frequently incomplete. The paper referral letter is produced under time pressure and commonly omits recent laboratory results, current medication dosages, allergy information, and imaging findings. Questionnaire data confirmed that 81% of clinical staff receive incomplete referral information at least once per week.

The process is slow. Observation data showed that the average time from referral decision to information availability at the receiving hospital was between 45 minutes and 3 hours. In emergency cases this delay directly compromises patient safety.

There is no feedback mechanism. Once a patient leaves the originating hospital with a referral letter, the referring clinician receives no confirmation of whether the patient arrived, whether they were admitted, or what treatment decision was made. This breaks the continuity of care.

3.7.2 Proposed System Improvements

The proposed platform addresses each of the existing system's limitations directly. Table 3.5 summarises the comparison between the existing system and the proposed platform.

Table 3.5: Comparison between existing system and proposed platform

Dimension	Existing system	Proposed platform
Data transfer method	Paper referral letter, printed lab results, CDs	Real-time electronic transfer via secure API
Data completeness	Dependent on referring clinician's time and attention	System enforces mandatory data fields (FR05)
Transfer time	45 minutes to 3 hours	Under 5 seconds (NFR01)
Access to prior records	Not available to receiving hospital	Full patient history accessible to authorized clinician
Security	Physical document security only	JWT authentication, TLS encryption, RBAC, audit logging
Referral status tracking	None	Real-time status updates visible on dashboard (FR07)
Continuity of care	Broken at point of referral	Maintained through shared access to patient records
Scalability	Cannot be extended	Architecture will support up to more than 10 hospital nodes

Where: NFR01 means Non functional requirement number one in Non functional requirement table and FR01 means Functional requirement number one in Functional requirement table.

3.8 System Diagrams

3.8.1 System Architecture Diagram

The system architecture follows a six-tier layered model. The tiers from top to bottom are: the user tier (referring and receiving clinicians), the presentation tier (clinician web interface), the security tier (JWT authentication and RBAC), the exchange tier (central interoperability API), the data tier (CHUK and KFH hospital databases), and the audit tier (transaction logging for both institutions). The architecture ensures that no direct connection exists between the two hospital databases, all data exchange passes through the central exchange layer, which validates, maps, and routes data according to standardised FHIR-aligned field definitions.

System Architecture Diagram

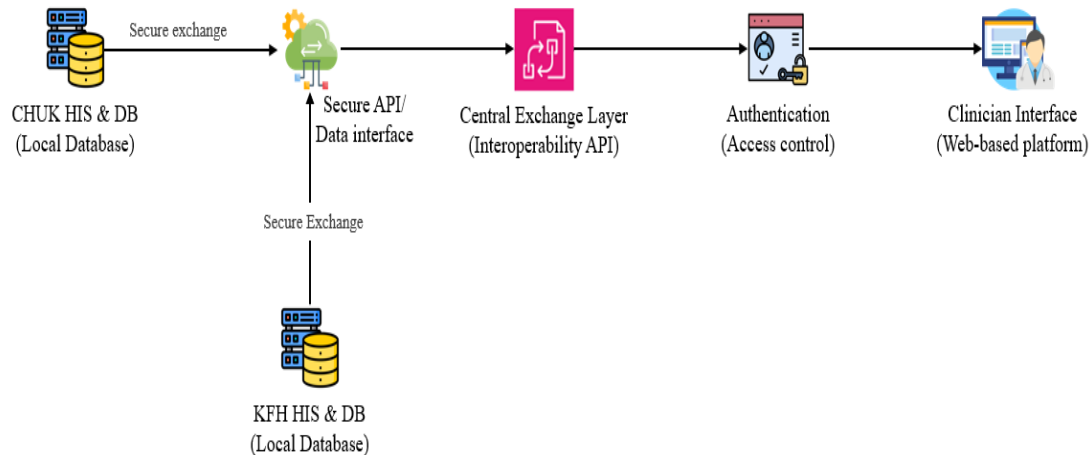


Figure 3.5.1 System Architecture Diagram

3.8.3 Data Flow Diagram (Level 0)

The Level 0 DFD illustrates the complete flow of patient data through the platform during a referral event. The referring clinician at CHUK authenticates through the central exchange system and creates a referral record. The exchange system validates the request, checks permissions via the RBAC module, stores the referral in the central referral database, and sends a notification to the receiving clinician at King Faisal Hospital. The receiving clinician authenticates, views the referral, and requests the patient's full records. The exchange system retrieves the relevant records from CHUK's database through the secure API connection and delivers them to the receiving clinician's interface. Every step in this flow is recorded in the audit logs of both institutions.

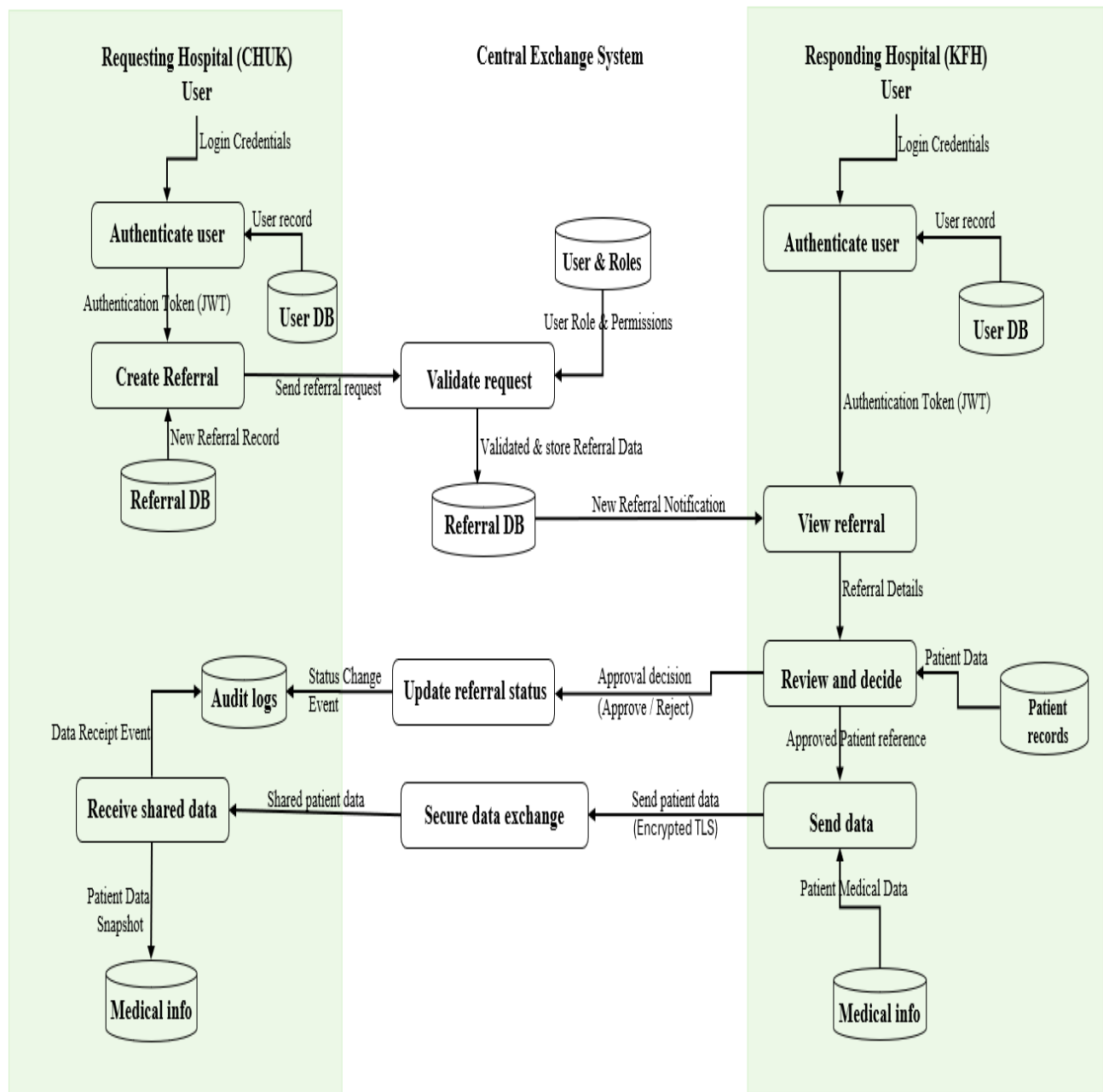


Figure 3.5.3 Data Flow Diagram (Level 0)

REFERENCES

- [1] Rwanda Biomedical Centre, "Rwanda Digital Health Strategic Plan 2020–2024," Ministry of Health, Rwanda, 2020.
- [2] Healthcare Information and Management Systems Society (HIMSS), "Health Information Exchange (HIE) in Healthcare," HIMSS, 2022.
- [3] Office of the National Coordinator for Health Information Technology (ONC), "Connecting Health and Care for the Nation: A Shared Nationwide Interoperability Roadmap," ONC, Washington DC, 2018.
- [4] T. Benson and G. Grieve, *Principles of Health Interoperability: SNOMED CT, HL7 and FHIR*, 3rd ed. Springer, 2016.
- [5] R. H. Dolin et al., "HL7 Clinical Document Architecture, Release 2," *Journal of the American Medical Informatics Association*, vol. 13, no. 1, pp. 30–39, 2006.
- [6] A. K. Jha et al., "Use of Electronic Health Records in U.S. Hospitals," *New England Journal of Medicine*, vol. 360, no. 16, pp. 1628–1638, 2009.
- [7] B. Blobel, "Comparing Approaches for Advanced eHealth Interoperability," *Studies in Health Technology and Informatics*, vol. 192, pp. 107–111, 2013.
- [8] Centers for Medicare and Medicaid Services, "Interoperability and Patient Access Final Rule," CMS, 2020.
- [9] eHealth Exchange, "About the eHealth Exchange Network," The Sequoia Project, 2023.
- [10] J. Adler-Milstein and A. K. Jha, "Health Information Exchange Among U.S. Hospitals: Who's In, Who's Out, and Why?" *Healthcare*, vol. 5, no. 1–2, pp. 26–32, 2017.
- [11] J. R. Vest and L. D. Gamm, "Health Information Exchange: Persistent Challenges and New Strategies," *Journal of the American Medical Informatics Association*, vol. 17, no. 3, pp. 288–294, 2010.
- [12] J. Walker et al., "The Value of Health Care Information Exchange and Interoperability," *Health Affairs*, vol. 24, pp. W5-10–W5-18, 2005.
- [13] G. J. Kuperman, "Health-Information Exchange: Why Are We Doing It, and What Are We Doing?" *Journal of the American Medical Informatics Association*, vol. 18, no. 5, pp. 678–682, 2011.
- [14] G. J. Kuperman, "Health-Information Exchange: Why Are We Doing It, and What Are We Doing?" *Journal of the American Medical Informatics Association*, vol. 18, no. 5, pp. 678–682, 2011.
- [15] T. Benson and G. Grieve, *Principles of Health Interoperability: SNOMED CT, HL7 and FHIR*, 3rd ed. Springer, 2016.